

O9V - FASTWIND model diagnostics

Stellar & wind parameters

Stellar

$T_{\text{eff}} = 32,882 \text{ K}$
 $\log g = 3.92$
 $R_* = 7.53 R_{\odot}$
 $R_{\text{max}} = 120 R_{\odot}$
 $T_{\text{min}}/T_{\text{eff}} = 0.6$

Wind

$\dot{M} = 6.6 \times 10^{-8} M_{\odot} \text{ yr}^{-1}$
 $v_{\infty} = 2300 \text{ km s}^{-1}$
 $\beta = 0.8$
 $v_{\text{min}} = 0.1 \text{ km s}^{-1}$
 $v_{\text{trans}} = 0.1 a_s$

Atmosphere

$v_{\text{turb, model}} = 10 \text{ km s}^{-1}$
 $\xi_{\text{He}} = 2$
 $f_{\text{cl, max}} = 10$
 $v_{\text{cl, start}}/v_{\infty} = 0.05$
 $v_{\text{cl, max}}/v_{\infty} = 0.1$

Derived parameters

$\log(L/L_{\odot}) = 4.776$

$M_{\text{spec}} = 17.20 M_{\odot}$

$\log Q = -13.536$

Abundances

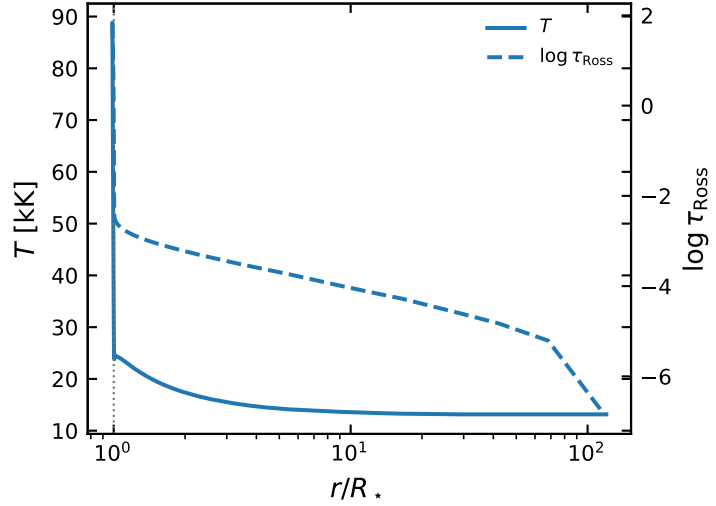
Global

$Z = 1 Z_{\odot}$
 $N_{\text{He}} = 0.11$

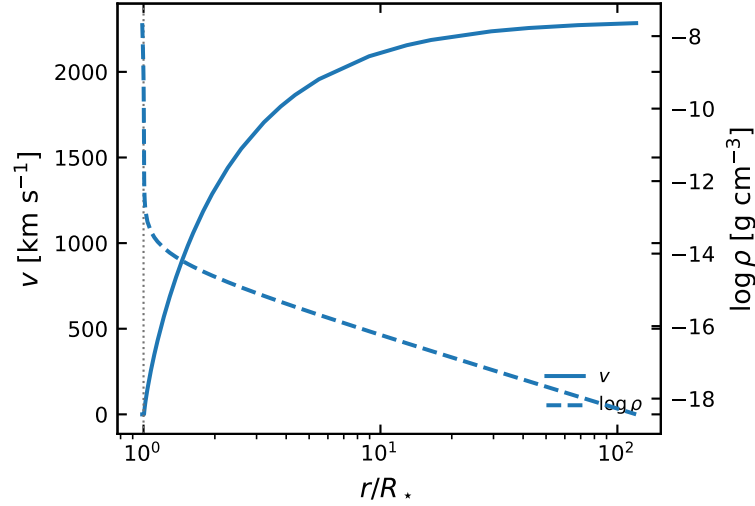
Individual

$\epsilon_{\text{C}} = 8.33$
 $\epsilon_{\text{N}} = 8.46$
 $\epsilon_{\text{O}} = 8.59$

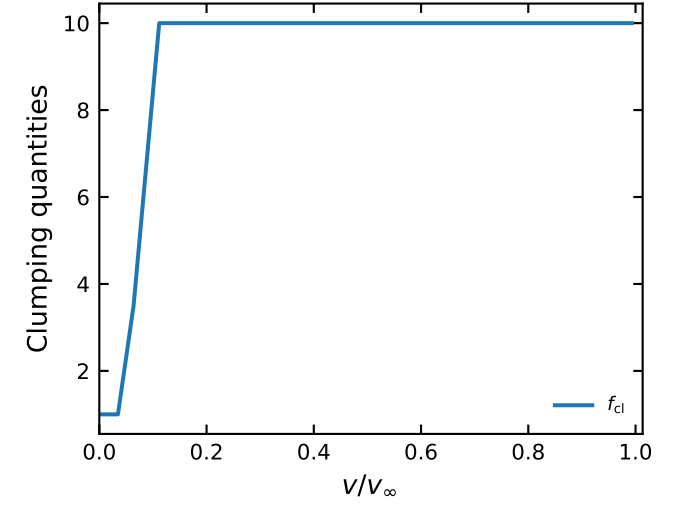
Thermal / optical-depth structure



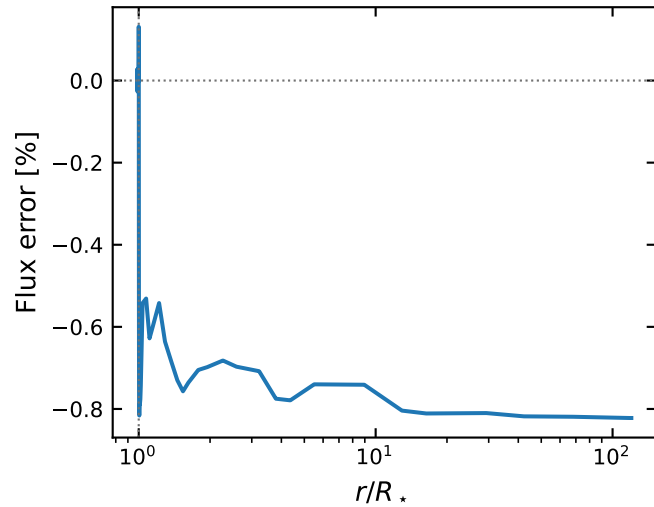
Velocity / density structure



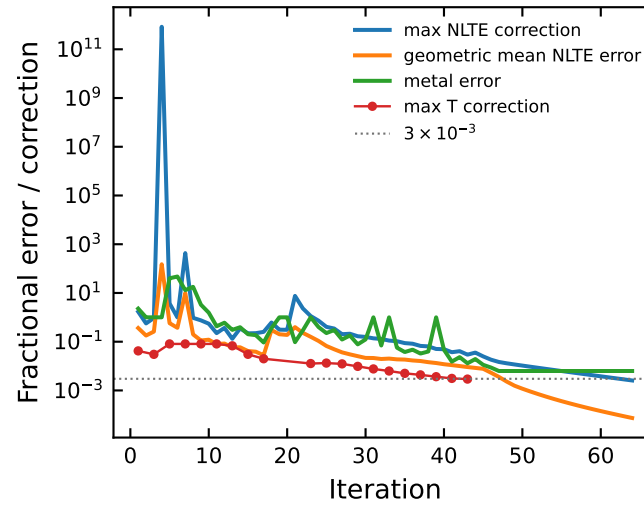
Clumping stratification



Flux conservation (max $|\Delta F/F| = 0.82\%$)



Convergence history



Radiative acceleration

